

1. Using the properties of linear transformations.

$$\text{a. } T \left(2 \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} \right) = 2T \left(\begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} \right) = 2 \begin{bmatrix} 3 \\ 0 \end{bmatrix} = \begin{bmatrix} 6 \\ 0 \end{bmatrix}$$

$$\text{b. } T \left(\begin{bmatrix} 3 \\ 6 \\ 3 \end{bmatrix} \right) = 3T \left(\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \right) = 3 \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -3 \\ 6 \end{bmatrix}$$

$$\text{c. } T \left(\begin{bmatrix} -1 \\ 3 \\ 0 \end{bmatrix} \right) = T \left(\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \right) - T \left(\begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} \right) = \begin{bmatrix} -1 \\ 2 \end{bmatrix} - \begin{bmatrix} 3 \\ 0 \end{bmatrix} = \begin{bmatrix} -4 \\ 2 \end{bmatrix}$$

2. Note that $x_1 + 2x_2 + x_3 = (1, 2, 1) \cdot (x_1, x_2, x_3)$ and $3x_1 - x_2 - x_3 = (3, -1, -1) \cdot (x_1, x_2, x_3)$. Also note that $A \in \mathcal{M}_{2 \times 3}$ so that $A\mathbf{x} \in \mathcal{M}_{2 \times 1}$. So

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & -1 & -1 \end{bmatrix}$$

Alternatively, $T \left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$, $T \left(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$, and $T \left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ gives the 1st, 2nd, and 3rd column of A respectively.

3. a. The 1st column is $T \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$.

For the 2nd column $T \left(\begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) = T \left(\begin{bmatrix} 2 \\ 1 \end{bmatrix} \right) - T \left(\begin{bmatrix} 2 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} -1 \\ 1 \end{bmatrix} - \begin{bmatrix} 4 \\ -6 \end{bmatrix} = \begin{bmatrix} -5 \\ 7 \end{bmatrix}$. So

$$A = \begin{bmatrix} 2 & -5 \\ -3 & 7 \end{bmatrix}$$

- b. The 2nd column is $T \left(\begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ -3 \end{bmatrix}$.

The 1st column is $T \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) = \frac{1}{2} \left[T \left(\begin{bmatrix} 2 \\ 1 \end{bmatrix} \right) - T \left(\begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) \right] = \frac{1}{2} \left[\begin{bmatrix} 5 \\ 3 \end{bmatrix} - \begin{bmatrix} 1 \\ -3 \end{bmatrix} \right] = \frac{1}{2} \begin{bmatrix} 4 \\ 6 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$. So

$$A = \begin{bmatrix} 2 & 1 \\ 3 & -3 \end{bmatrix}$$

- c. $2T \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) = T \left(\begin{bmatrix} 2 \\ 0 \end{bmatrix} \right) = T \left(\begin{bmatrix} 1 \\ 1 \end{bmatrix} \right) + T \left(\begin{bmatrix} 1 \\ -1 \end{bmatrix} \right) = \begin{bmatrix} 3 \\ 3 \end{bmatrix} + \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$; so the 1st column is $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ (divide by 2).

$2T \left(\begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) = T \left(\begin{bmatrix} 1 \\ 1 \end{bmatrix} \right) - T \left(\begin{bmatrix} 1 \\ -1 \end{bmatrix} \right) = \begin{bmatrix} 3 \\ 3 \end{bmatrix} - \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$; so the 2nd column is $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$. Thus

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$$

4. Recall that to prove that T is a linear transformation it suffices to find a standard matrix, $A \in \mathcal{M}_{m \times n}$ for $\mu_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$. To prove that it is not a standard matrix you need to show that it fails at least one of the properties.

- a. T is not a linear transformation.

$$\text{Proof. } -1 \cdot T \left(\begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) = - \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ -1 \end{bmatrix} \neq T \left(\begin{bmatrix} 0 \\ -1 \end{bmatrix} \right) = \begin{bmatrix} -2 \\ 1 \end{bmatrix}. \quad \square$$

- b. T is a linear transformation.

Proof. $T \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $T \left(\begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$ gives the 1st and 2nd column respectively of $A = \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$, and $\begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 + 2x_2 \\ 0 \end{bmatrix}$ so $\mu_A = T$ which shows that T is linear. $\square$

6. a. $T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and $T\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} -1 \\ 0 \end{bmatrix} \Rightarrow A_T = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

It is a reflection across $x_1 + x_2 = 0 \Rightarrow x_1 = -x_2$. So then $S\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$ and $S\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} -1 \\ 0 \end{bmatrix} \Rightarrow A_S = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$.

b. $T \circ S$ notates the composition function, i.e., the points are mapped first by S and then by T . So $(T \circ S)\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = T\left(\begin{bmatrix} 0 \\ -1 \end{bmatrix}\right) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $(T \circ S)\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = T\left(\begin{bmatrix} -1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$. So the standard matrix is $A_{T \circ S} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$. Note that $TS = A_{T \circ S}$.

c. $(S \circ T)\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = S\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$ and $(S \circ T)\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = S\left(\begin{bmatrix} -1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. So the standard matrix is $A_{S \circ T} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$. Note that $ST = A_{S \circ T}$.